

THE UNIFIED BLOCKWAVE ACTION

A WORLDLINE-FIRST REFORMULATION OF THE SAT FRAMEWORK

ARCHITECTURAL TRANSITION

The present iteration of the Scalar–Angular–Twist (SAT) framework replaces field-based Effective Field Theory constructions with a mathematically closed, worldline-first architecture. EFT representations are regarded as auxiliary and are admissible only when explicitly mapped into the four-dimensional Hyperhelical (4DHH) ontology.

The SAT.4DHH Lagrangian serves as the primary dynamical object. It embeds into the Unit Cell Particle-Only Lagrangian, which itself embeds into the Full Four-Dimensional Unit Cell Asymmetry Lagrangian. This nested hierarchy eliminates the point-particle approximation in favor of a resolved four-dimensional block of oscillatory worldlines: the *Zottenwelt*.

FOUNDATIONAL ONTOLOGY

The Zottenwelt is a smooth four-manifold M populated by one-dimensional physical worldlines termed filaments. Each filament is parametrized by the hyperhelical trajectory

$$\gamma^\mu(\lambda) = \left(\lambda, A_x \sin(k_x \lambda + \phi_x), A_y \sin(k_y \lambda + \phi_y), A_z \sin(k_z \lambda + \phi_z) \right), \quad (1)$$

where λ denotes internal time and (A, k, ϕ) represent amplitude, pitch, and phase parameters.

Particles are not primitive objects. They arise as zero-dimensional intersection loci of progenitor hyperspheres within a space-filling 24-cell lattice. A quadruple intersection,

$$S^3 \cap S^3 \cap S^3 \cap S^3,$$

defines a single 0D particle event.

TRIATTIC CORRECTION AND PARAMETER COLLAPSE

The Triattic Correction asserts that physical attributes—spin, mass, charge—are not localized at point-instants but are integrated geometric invariants of a filament’s coiling history. Property realization occurs only when the resolving three-dimensional time surface Σ_t intersects the four-dimensional filament lattice.

The framework enforces a Zero-Parameter Economy. All dimensionless ratios derive from internal filament topology. A single dimensional anchor (e.g., filament tension T or base frequency) determines scale, while all remaining ratios follow from the Filament Angle Constant,

$$B \approx 0.2387 \text{ rad}, \tag{2}$$

interpreted as the per-dimension distortion generated when symmetric four-dimensional dynamics project into three dimensions.

EMERGENCE OF FORCES AND CHIRALITY

Forces are structural consequences of filament interlocking and rotational dynamics.

MASS AND INERTIA

Mass is identified with Projective Resistance R encountered by the resolving surface Σ_t as it traverses hyperhelical excitations. Under the Proportional Mass Law,

$$m \propto Q,$$

where topological complexity Q forces filaments into high misalignment angles θ_4 relative to the time-flow vector u^μ .

ELECTROWEAK DYNAMICS

Electroweak behavior arises from chiral motion of the unit cell. The rotational term Ω encodes relative angular momentum among the 24 progenitor hyperspheres. Static configurations are mirror-degenerate; relative rotation breaks this symmetry and separates intersections into left-, right-, and mixed-handed subclasses.

GAUGE UNIFICATION

Absolute interaction strengths scale with linking density ρ_{link} and filament scale ℓ_f , which anchors the transverse weave at

$$\ell_f \approx 0.7937 \text{ fm.}$$

CONSTRAINT RIGIDITY AND FALSIFIABILITY

The framework is governed by strict, non-adjustable structural constraints.

- **UV Finiteness Lock** ($Q \leq 3$). No stable ground state may contain more than three filaments. Observation of a stable $Q = 4$ tetraquark would falsify the core filament architecture.
- **Optical Quantization.** An achromatic geometric phase shift of approximately 0.246 radians is predicted for light traversing a θ_4 domain wall.
- **Gravitational Signature.** The one-loop effective action fixes the higher-curvature coefficient

$$\beta = \frac{2}{6\pi} \approx 0.106,$$

establishing a non-adjustable R^2 contribution.

- **Holonomy Locking.** Internal lattice closure enforces the Dirac CP phase at precisely 270° , interpreted as a quarter-turn holonomy condition.